

Attention Mechanisms in Sequence Models (study notes)

Problem: recurrent models process sequences step by step, which makes them slow to train and makes it hard for information from early in a long sequence to influence predictions much later on. This is often called the long range dependency problem.

The core idea covered in these notes is self-attention: instead of processing tokens strictly in order, every token attends directly to every other token in the sequence in a single step. Each token produces a query, a key, and a value vector. The attention weight between two tokens comes from the dot product of one token's query with another's key, scaled and passed through softmax, and the output is a weighted sum of value vectors.

Multi head attention runs several attention operations in parallel with different learned projections, then concatenates the results. The intuition is that different heads can specialize, one might track subject verb agreement, another might track coreference, without anyone hand designing that split.

Method summary: stack multi head self attention layers with feed forward layers and residual connections, no recurrence at all. Position information is added explicitly since attention itself has no notion of order.

Result summary: this architecture trained faster than recurrent alternatives on the tasks tested and reached better quality, largely because the full sequence can be processed in parallel instead of one step at a time.